

Unit 6, Lesson 3: Preserving Distance



Benchmarks

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.



Learning Targets

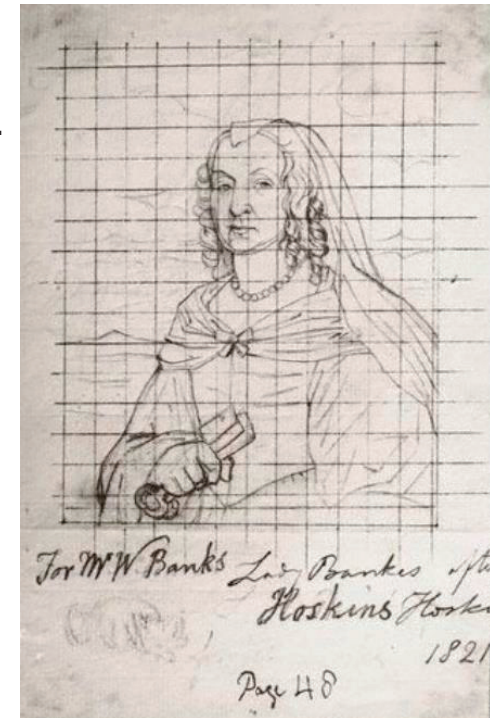
- ☐ I can use a description of a sequence of transformations to determine if the transformation does or does not preserve distance.
- ☐ I can draw a sequence of transformations to determine if the transformation does or does not preserve distance.
- ☐ I can identify transformations that do or do not preserve distance.



6.3.1 Warm-Up: Notice and Wonder

Artist John Hoskins sketched a drawing of Mary Hawtrey, also known as Lady Mary Bankes, in 1821. Henry Pierce Bone then used the sketch to create an enamel miniature painting.

Following the death of her husband, Lady Mary Bankes successfully defended the Corfe Castle in Dorset, England during a siege in 1643. Unfortunately, during a second siege in 1646 the castle was destroyed.



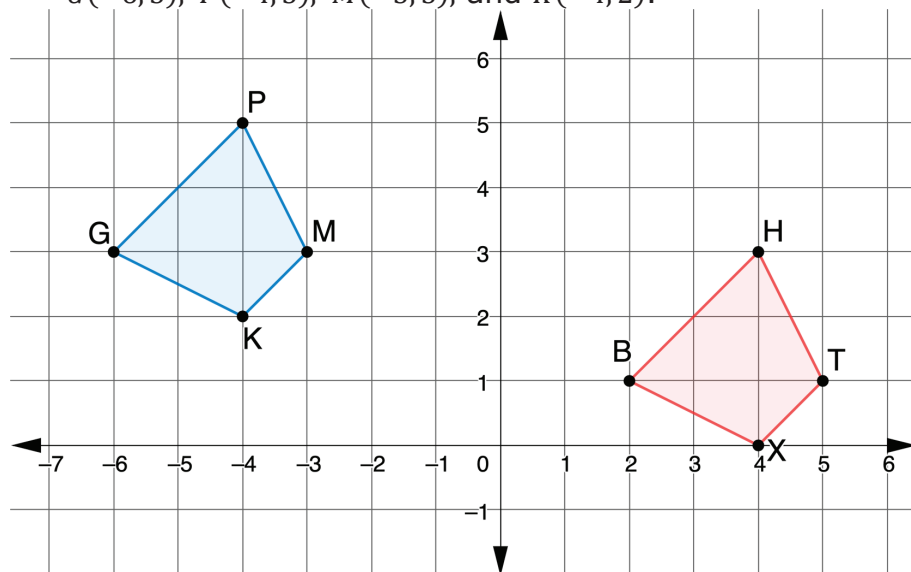
1. What do you notice about the sketch of Lady Mary Bankes?
2. What do you wonder about the sketch of Lady Mary Bankes or her story?



6.3.2 Exploration: Sequence of Transformations

With your partner, answer the following questions.

1. $BHTX$ with vertices $B(2, 1)$, $H(4, 3)$, $T(5, 1)$, and $X(4, 0)$ is the result of a transformation of $GPMK$ with vertices $G(-6, 3)$, $P(-4, 5)$, $M(-3, 3)$, and $K(-4, 2)$.



- a. What transformation occurred to $GPMK$ that resulted in $BHTX$?
- b. Use patty paper to trace $GPMK$, then place $GPMK$ onto $BHTX$.
- c. What do you notice about the distances between the vertices of corresponding sides?

- d. What can be concluded about the lengths of the corresponding sides after this transformation occurs?

2. The coordinates of $RWTY$ and $R'W'T'Y'$ are shown in the table, where $RWTY$ is the preimage.

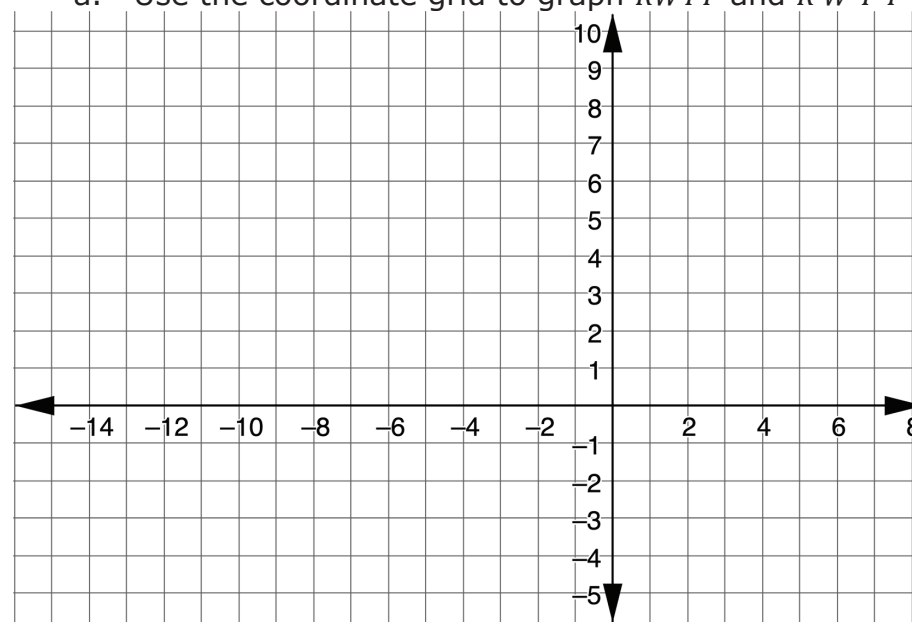
$RWTY$

Vertex	x	y
R	-6	8
W	-4	6
T	2	4
Y	-6	2

$R'W'T'Y'$

Vertex	x	y
R'	-2	-4
W'	0	-2
T'	2	4
Y'	4	-4

- a. Use the coordinate grid to graph $RWTY$ and $R'W'T'Y'$.



b. What type of transformation was applied to the preimage, $RWTY$?

c. How could the coordinates of the preimage and image be used to show that the transformation preserved side length?

3. The coordinates of $MPRD$ and $SGTW$ are shown in the table, where $MPRD$ is the preimage.

MPRD

Vertex	x	y
M	8	-7
P	9	-4
R	4	-2
D	1	-5

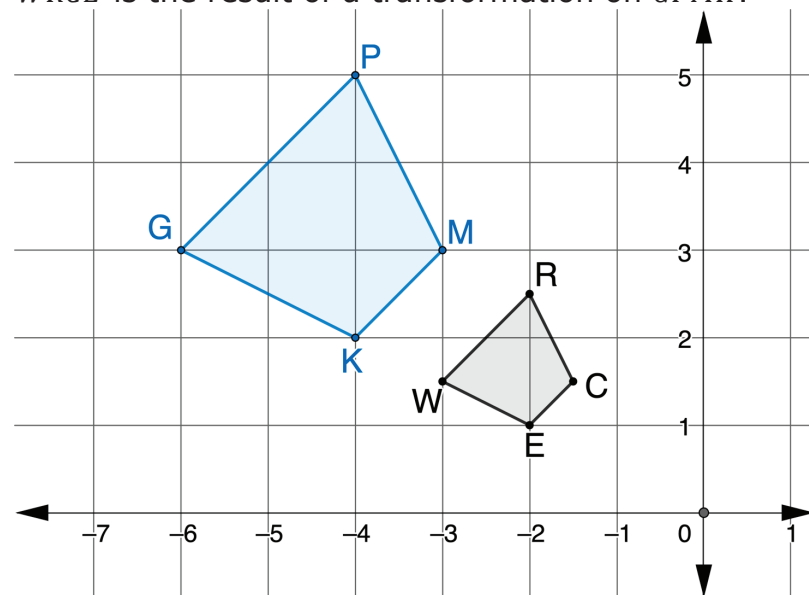
SGTW

Vertex	x	y
S	8	7
G	9	4
T	4	2
W	1	5

a. What transformation occurred from the preimage to the image?

b. Explain how the coordinates of the preimage and image show that the transformation preserved distance.

4. $WRCE$ is the result of a transformation on $GPMK$.



a. What transformation occurred to $GPMK$ that resulted in $WRCE$?

b. What can be concluded about the lengths of each corresponding side after this transformation occurs?



6.3.3 Bring It Together: Preserving Distance

1. Complete the statements based on the Exploration.

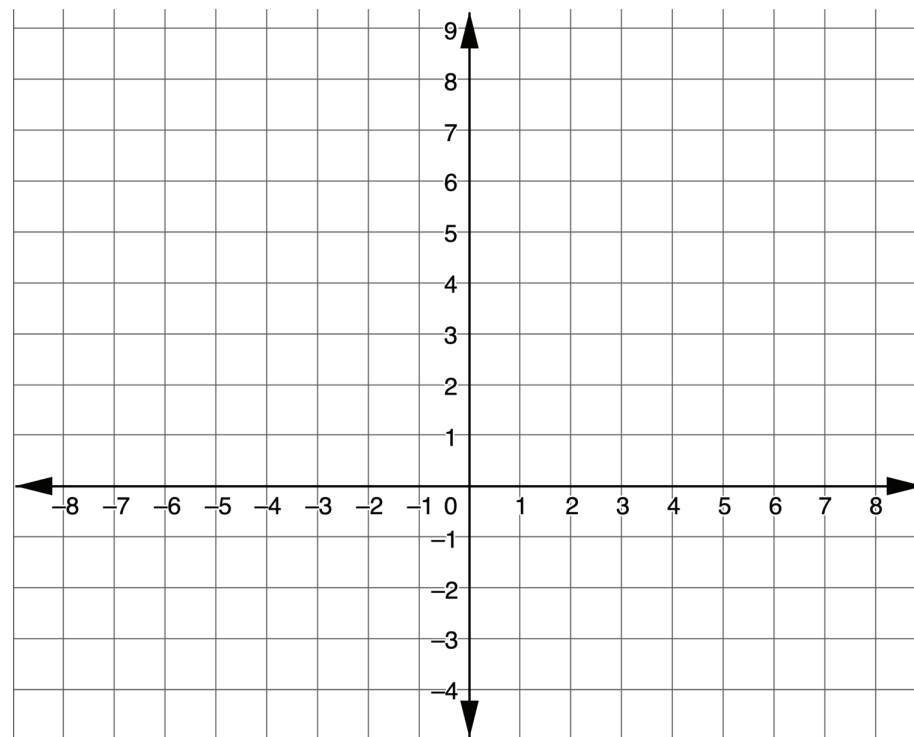
- a. $BHTX$ was created by
☐ reflection.
☐ translation.
☐ rotation.
☐ dilation.
 This transformation
☐ does
☐ does not
 preserve distance between the vertices.
- b. $R'W'T'Y'$ was created by
☐ reflection.
☐ translation.
☐ rotation.
☐ dilation.
 This transformation
☐ does
☐ does not
 preserve distance between the vertices.
- c. $SGTW$ was created by
☐ reflection.
☐ translation.
☐ rotation.
☐ dilation.
 This transformation
☐ does
☐ does not
 preserve distance between the vertices.
- d. $WRCE$ was created by
☐ reflection.
☐ translation.
☐ rotation.
☐ dilation.
 This transformation
☐ does
☐ does not
 preserve distance between the vertices.



6.3.4 Guided Instruction: Sequence of Transformations and Distance Preservation

1. $GPMK$ with vertices $G(-6, 3)$, $P(-4, 5)$, $M(-3, 3)$, and $K(-4, 2)$ is reflected over $x = -2$, then translated right 2 units and up 1 unit to create $ZFVN$, respectively.

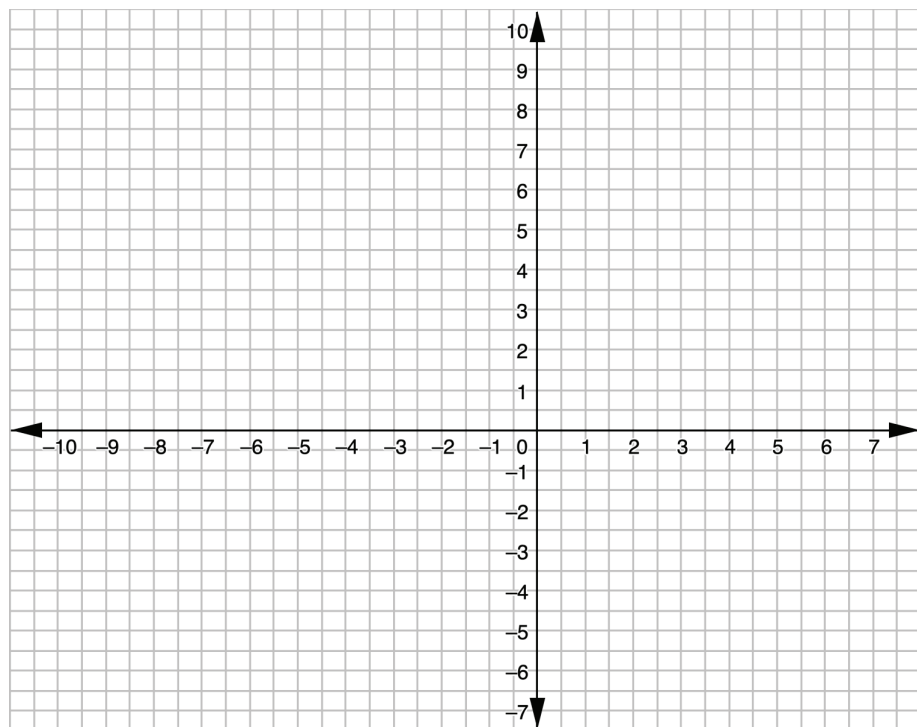
a. Draw the preimage $GPMK$ and the image $ZFVN$.



b. Use patty paper or the coordinates to determine if the corresponding sides of $GPMK$ and $ZFVN$ are congruent.

2. $THNK$ with vertices $T(-9,3)$, $H(-7.5,6)$, $N(-3,3)$, and $K(-3,-6)$ is rotated 90° clockwise about the origin, then dilated using the scale factor of $\frac{1}{3}$ centered at the origin to create $T''H''N''K''$, respectively.

a. Draw the preimage $THNK$ and the image $T''H''N''K''$.



b. Determine if the corresponding sides of $THNK$ and $T''H''N''K''$ are congruent.

- c. Explain how you could use the description of a sequence of transformations to determine if the transformation does or does not preserve distance.

d. How would your response differ if you were given the coordinates of a sequence of transformations?



6.3.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the lesson that I knew were ...	Something I wondered during the lesson was ...	Something I learned during the lesson was ...



Unit 6, Lesson 4: Sequence of Transformations



Benchmarks

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.



Learning Targets

- ☐ I can represent a sequence of transformations that will map the preimage onto the image.
- ☐ I can describe a sequence of transformations that will map the preimage onto the image.
- ☐ I can identify transformations that do or do not preserve distance.
- ☐ I can use a description of a sequence of transformations to determine if the transformation does or does not preserve distance.